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DCT-Enhanced Lithography Hotspot Detection

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***Abstract*— Lithography hotspot detectors based on discrete cosine transform (DCT) features are common in the literature, but the benefit of a frequency-domain input to a small convolutional neural network (CNN) is rarely isolated under controlled conditions. We test it on the ICCAD-12 benchmarks (B1–B5) with a 12,873-parameter CNN, a fixed stratified 80/20 train/validation split (seed 42), and three training seeds (42, 123, 2024). The official test partition is held out and not used for any result reported here. Factors examined are DCT coefficient count ($K = 64$ to 16,384, plus the full $224\times224 = 50{,}176$-coefficient transform), coefficient ordering (zig-zag versus row-major), an initial block-DCT screening on B1, and multi-seed robustness. Balanced accuracy (BA) on the validation split is the primary metric. Truncated DCT with zig-zag selection and $K = 4096$ reached a benchmark-level mean BA of 55.94% against 87.05% for the spatial baseline ($+31.11$ percentage points for spatial). A paired t-test over the five benchmark means gave $p = 0.0107$, whereas the exact Wilcoxon test gave $p = 0.0625$, so the evidence is suggestive rather than conclusive. Eleven of the fifteen $K = 4096$ runs were constant classifiers, and B4 ranged from 50.0% to 91.1% across seeds. Coefficient count showed no monotonic trend, full-image DCT behaved differently from truncated DCT, and the row-major variant did not exceed chance. Within the tested design space, truncated DCT did not provide a robust improvement over the spatial representation; the study does not establish that frequency-domain inputs are unsuitable in general.**

***Index Terms*— Lithography hotspot detection, discrete cosine transform, frequency-domain representation, convolutional neural network, balanced accuracy, ICCAD-12, robustness.**

I. Introduction

Layout patterns that print poorly after sub-wavelength lithography, known as hotspots, must be found before mask synthesis. Full lithography simulation is accurate but expensive, so learned detectors operating on rasterized layout clips have become the usual alternative [3, 4, 12]. A recurring design choice concerns the input representation. DCT coefficients are attractive because spacing and edge density map onto spatial frequency, and truncating high-frequency terms shrinks the input. Block-wise DCT features were used in an influential CNN detector [2, 3], and DCT-domain inputs have also been explored for general image classification [9].

Whether the transform helps a small CNN, however, depends on choices that are seldom varied together: how many coefficients are kept, in what order, over what block size, and how stable the outcome is across random initializations. This paper treats the question empirically: *does a DCT/frequency-domain representation improve hotspot detection relative to the spatial baseline, and how sensitive is the outcome to coefficient selection, coefficient ordering, block size and random initialization?*

The contributions are (i) a controlled comparison on all five ICCAD-12 benchmarks under one validation protocol, (ii) coefficient-count and ordering ablations with the full-image DCT kept separate from truncated variants, (iii) an initial block-DCT screening, and (iv) a three-seed robustness analysis with a statistical comparison designed to avoid pseudoreplication. The main finding is negative under the evaluated configuration, and it is reported as such.

II. Related Work

Early hotspot detectors combined pattern matching with classical learning, for example refined pattern identification [7] and PCA with a support vector machine [8]. The ICCAD-2012 contest benchmark [1] supplied a shared testbed. CNN detectors followed: Shin and Lee applied deep CNNs to layout images [4], Matsunawa *et al.* learned layout features with a deep network [5], and Yang *et al.* addressed class imbalance through biased learning [6].

The closest precedent is the feature-tensor method of Yang *et al.* [2, 3]. It partitions a layout clip into sub-blocks, applies a DCT to each block, flattens the coefficients in zig-zag order, discards high frequencies, and stacks the retained coefficients as channels of a tensor fed to a CNN. Outside lithography, Xu *et al.* fed DCT coefficients to standard image-classification networks and selected frequency channels, reporting gains on ImageNet [9]. The spatial baseline used here follows the lightweight 12,873-parameter CNN of Verma *et al.* [12]. Section IX compares these works with the present design.

III. Dataset and Experimental Setup

Experiments use the ICCAD-12 benchmarks B1–B5 [1]. Each benchmark provides training and official test partitions of grayscale PNG clips of roughly 1200×1200 pixels (Table I). Each official training partition is split once into a stratified 80% training subset and 20% validation subset with split seed 42; the same split is used for every method. Training seeds are 42, 123 and 2024. The official test partitions were not touched during training or validation.

Scope of reported metrics. The supplied run-level file (metrics.csv) records the validation-split size for each run (88, 1,092, 1,111, 910 and 549 clips for B1–B5), and the hotspot and non-hotspot counts in every confusion matrix match the 20% validation partition exactly. All numerical results below therefore refer to the validation split. No official-test metrics were available for this paper, and none are claimed. Because training used

Table I. Dataset distribution of the ICCAD-12 benchmarks. The 20% validation split (seed 42) contains 20/68, 35/1057, 182/929, 19/891 and 5/544 HS/NHS clips for B1–B5.

Bench.	Train HS	Train NHS	Train total	Test HS	Test NHS	Test total
B1	99	340	439	226	4,679	4,905
B2	174	5,285	5,459	498	41,298	41,796
B3	909	4,643	5,552	1,808	46,333	48,141
B4	95	4,452	4,547	177	31,890	32,067
B5	26	2,716	2,742	41	19,327	19,368

Table II. Baseline CNN architecture (reconstructed). *Valid padding; 7×7×12 = 588. Total 12,873 = 12,825 trainable + 48 non-trainable BatchNorm statistics.

Layer	Block/output	Params
Input	224×224×3	0
Conv 3×3, 12, ELU	Block 1	336
Conv 3×3, 12, ELU	Block 1	1,308
Conv 3×3, 12, linear	Block 1	1,308
BN, ELU, MaxPool 2×2	Block 1	48
MaxPool 5×5	–	0
3 Conv (12), BN, ELU, MaxPool	Block 2	3,972
Flatten, Dropout 0.3	588 units*	0
Dense(10), ReLU	–	5,890
Dense(1), sigmoid	–	11
Total		**12,873**

a fixed number of epochs with no checkpoint selection, the validation split was not used for model selection; it nevertheless remains a small, in-distribution evaluation set, and one hotspot on B5 (five in total) shifts BA by 10 percentage points.

Data integrity checks. All 62 rows of metrics.csv satisfy TP+FN = validation hotspots, TN+FP = validation non-hotspots, BA = (sensitivity+specificity)/2, and sensitivity/specificity recomputed from the counts. Two issues were found and not silently corrected: (a) the CSV mixes three column layouts, so rows were parsed by position within each layout; (b) the B1 full-DCT row appears twice with identical values and was counted once. The CSV has no row for spatial B4 with seed 2024; its confusion matrix (TP 16, TN 761, FP 130, FN 3; BA 84.81%) was supplied by the authors from the experiment log and is marked as such in Tables VII and VIII. The block-DCT rows label both runs $K = 4096$ (the requested target); the actual retained counts (Table VI) come from the implementation configuration.

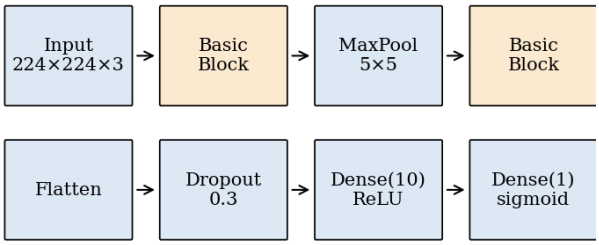
IV. Baseline CNN

Inputs are 224×224×3 tensors: the grayscale clip is resized to 224×224, scaled to $[0,1]$ and replicated over three channels. The network (Table II, Fig. 1) is Input → Basic Block → MaxPool 5×5 → Basic Block → Flatten → Dropout 0.3 → Dense(10, ReLU) → Dense(1, sigmoid), where a Basic Block is three 3×3 convolutions with 12 filters (ELU, ELU, linear), batch normalization, ELU and 2×2 max pooling. With valid padding the spatial size shrinks 224 → 218 in the first block, → 109 after pooling, → 21 after the 5×5 pool, → 15 in the second block and → 7 after the last pooling, so the flatten layer has 7×7×12 = 588 units. The model has 12,873 parameters (12,825 trainable plus 48 non-trainable BatchNorm statistics). Training used Nadam [11], binary cross-entropy, batch size 32, 5 epochs for B1 and 10 epochs for B2–B5, and a 0.5 decision threshold; no class weighting, resampling or threshold tuning was applied. The same network and recipe serve every representation, so only the input differs.

V. DCT-Based Representation

The clip is resized to 224×224, scaled to $[0,1]$, and a two-dimensional orthonormal DCT-II [10] is computed:

$$G(u,v) = \alpha(u)\alpha(v) \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} g(m,n) \cos \frac{\pi(2m+1)u}{2N} \cos \frac{\pi(2n+1)v}{2N}, \quad (1)$$



Basic Block: 3×[Conv 3×3, 12 filters] (ELU, ELU, linear) → BatchNorm → ELU → MaxPool 2×2. Total: 12,873 parameters

Fig. 1. Baseline CNN. The 12,873-parameter network takes a 224×224×3 input; each Basic Block contains three 3×3 convolutions with 12 filters.

with $N = 224$, $\alpha(0) = \sqrt{1/N}$ and $\alpha(k{>}0) = \sqrt{2/N}$. The full-image pipeline is DCT → (optional coefficient masking) → magnitude $|G|$ → $\log(1+|G|)$ → per-image min–max normalization → replication into three identical channels (Fig. 2).

Full-image DCT (R1) keeps all $224 \times 224 = 50{,}176$ coefficients. *Truncated DCT* keeps only K coefficients selected by (Z1) standard diagonal zig-zag order, which favors low frequencies, or (Z2) row-major order. The retained coefficients stay at their original positions in the 224×224 DCT matrix and all others are set to zero; they are not reshaped into a K-element vector. The CNN therefore always receives a 224×224×3 tensor, and K only controls how many coefficients carry information. The $K = 4096$ experiments are different from R1 and are never substituted for it. *Block-DCT* applies the DCT within non-overlapping blocks (16×16 and 56×56 tested), with a requested target of $K = 4096$. The target is not attainable for every block size: 16×16 gives 14×14 = 196 blocks with 20 coefficients each, i.e. 3,920 retained, whereas 56×56 gives 4×4 = 16 blocks with 256 each, i.e. exactly 4,096.

VI. Experimental Protocol

Balanced accuracy, $\mathrm{BA} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right)$, is the primary metric because it assigns 50% to any constant classifier and the classes are strongly imbalanced (hotspots are 0.21–4.61%

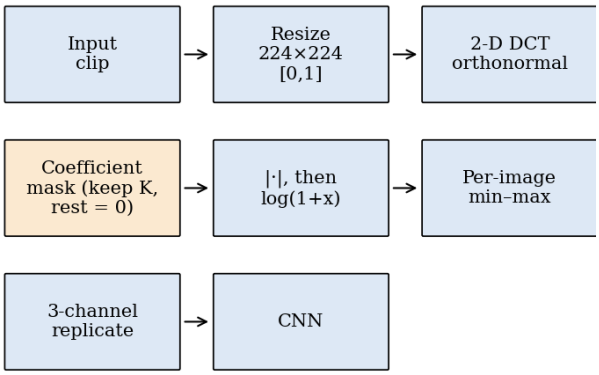


Fig. 2. DCT preprocessing. Full-image DCT (R1) retains all 50,176 coefficients; truncated variants retain K coefficients, e.g. $K = 4096$.

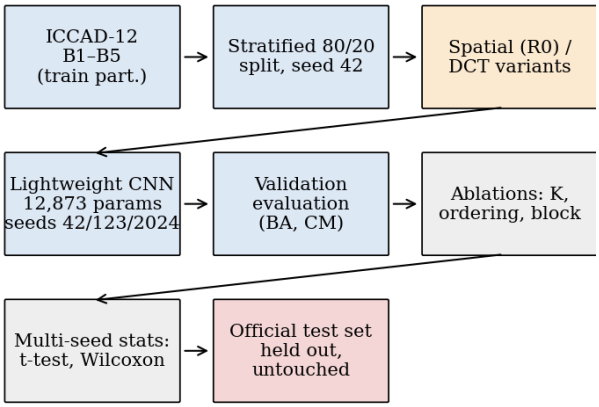


Fig. 3. Experimental methodology. Official test partitions remain untouched; all reported metrics come from the validation split.

of official test clips). Sensitivity, specificity, precision, F1 and full confusion matrices are taken from metrics.csv. Coefficient-count and ordering ablations use seed 42 only. The multi-seed experiment compares the spatial baseline (R0) with DCT zig-zag $K = 4096$ over seeds 42, 123 and 2024, with the split fixed. For significance testing the unit of analysis is the benchmark: per-benchmark means over seeds give five paired observations. The fifteen seed-level pairs are not treated as independent replicates, since seeds within a benchmark share data and split. A paired t-test, its 95% confidence interval and an exact Wilcoxon signed-rank test are reported.

Table III. Spatial baseline versus full-image DCT (R1, 50,176 coefficients): validation BA (%), seed 42.

Bench.	Spatial BA	R1 BA	Spatial TP/TN/FP/FN	R1 TP/TN/FP/FN
B1	80.00	50.00	12/68/0/8	0/68/0/20
B2	95.71	87.56	32/1057/0/3	35/794/263/0
B3	98.63	91.33	177/929/0/5	181/773/156/1
B4	83.42	80.91	13/877/14/6	12/879/12/7
B5	80.00	50.00	3/544/0/2	0/544/0/5
Mean	87.55	71.96		

Table IV. DCT coefficient-count ablation (zig-zag), validation BA (%), seed 42. "Full" = 50,176 coefficients.

Bench.	64	256	1024	4096	16384	Full
B1	50.00	50.00	50.00	50.00	50.00	50.00
B2	50.00	50.00	50.00	50.00	50.00	87.56
B3	50.00	76.96	50.00	50.00	50.00	91.33
B4	50.00	50.00	70.94	91.09	80.46	80.91
B5	50.00	77.43	50.00	50.00	50.00	50.00
Mean	50.00	60.88	54.19	58.22	56.09	71.96

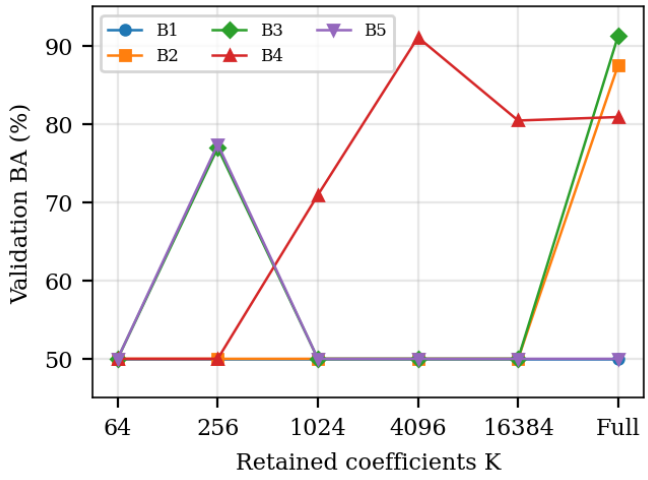


Fig. 4. Validation BA versus retained coefficients K for each benchmark (seed 42). "Full" denotes the 50,176-coefficient transform.

VII. Results

A. Spatial Baseline versus Full-Image DCT

Table III compares the spatial baseline with R1 (seed 42). R1 was competitive on B2–B4 but at chance on B1 and B5, where it predicted every clip as non-hotspot, and it remained below the spatial baseline on average.

B. Coefficient-Count Ablation

Truncated DCT with zig-zag selection gave no monotonic dependence on K (Table IV, Fig. 4). Most cells sit at exactly 50%, from all-negative or all-positive predictions; non-trivial results appeared only for B3 ($K = 256$), B4 ($K \geq 1024$, peaking at $K = 4096$) and B5 ($K = 256$). The best K differs per benchmark, and the full transform exceeded every truncated setting on B2 and B3 but not on B4 (91.09% at $K = 4096$) or B5.

Table V. Zig-zag versus row-major selection, $K = 4096$, validation BA (%), seed 42.

Bench.	Zig-zag	Row-major
B1	50.00	50.00
B2	50.00	50.00
B3	50.00	50.00
B4	91.09	50.00
B5	50.00	50.00
Mean	58.22	50.00

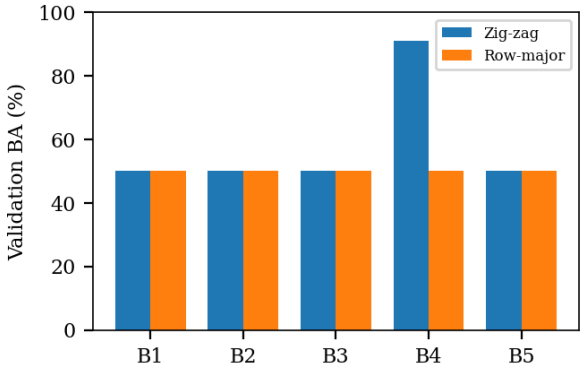


Fig. 5. Zig-zag versus row-major selection at $K = 4096$ (seed 42).

Table VI. Block-DCT screening on B1 (seed 42, initial screening only).

Block	Target K	Retained	BA (%)	TP/TN/FP/FN
16×16	4,096	3,920	50.00	0/68/0/20
56×56	4,096	4,096	50.00	0/68/0/20

C. Zig-Zag versus Row-Major Ordering

At $K = 4096$, row-major selection produced 50.00% on all five benchmarks, whereas zig-zag selection reached 91.09% on B4 only (Table V, Fig. 5), a mean difference of 8.22 pp. As this is a single-seed comparison and the zig-zag result rests on one benchmark, it indicates that outcomes depend on which coefficients are kept but does not establish a general ordering effect.

D. Block-DCT Screening

Initial B1 screening of 16×16 and 56×56 block-DCT configurations yielded 50% balanced accuracy with zero hotspot recall (Table VI, Fig. 6). Consequently, block-DCT was not pursued across all benchmarks under the available computational budget. This is a screening result on one benchmark and one seed, not a block-size ablation.

E. Multi-Seed Robustness

Table VII and Fig. 7 summarize the comparison. Across the five benchmark-level means the spatial baseline reached $87.05 \pm 9.87\%$ and the $K = 4096$ zig-zag DCT $55.94 \pm 10.97\%$. Of the fifteen DCT runs, ten predicted only non-hotspots, one only hotspots (B1, seed 123), and one (B3, seed 2024) found two of 182 hotspots; only B4 (seeds 42 and 2024) and B5 (seed 123) separated the classes. B4 ranged from 50.0% to 91.1%. The spatial baseline was also seed-dependent on some benchmarks (B3: 72.5% to 99.5%; B1: 62.5% to 80.0%), so seed sensitivity is not unique to the DCT input, although for DCT it was far more often accompanied by complete collapse. Fig. 8 shows seed-42 confusion matrices for B3 and B4, and Table VIII lists every matrix.

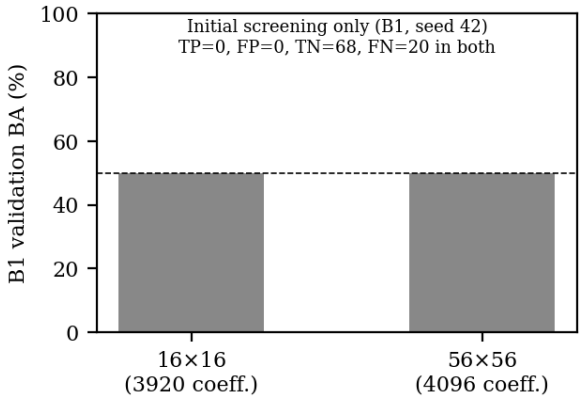


Fig. 6. Initial block-DCT screening on B1. Only two block sizes were tested; this is not a complete block-size study.

Table VII. Multi-seed validation BA (%): spatial versus DCT zig-zag $K = 4096$. *Sample SD of the five benchmark-level means. †Not in metrics.csv; taken from the authors' experiment log.

Bench.	Sp. 42	Sp. 123	Sp. 2024	Spatial	DCT 42	DCT 123	DCT 2024	DCT K=4096
B1	80.00	72.50	62.50	71.67±8.78	50.00	50.00	50.00	50.00±0.00
B2	95.71	99.57	100.00	98.43±2.36	50.00	50.00	50.00	50.00±0.00
B3	98.63	99.51	72.53	90.22±15.33	50.00	50.00	50.55	50.18±0.32
B4	83.42	86.56	84.81†	84.93±1.57	91.09	50.00	84.81	75.30±22.13
B5	80.00	100.00	90.00	90.00±10.00	50.00	62.68	50.00	54.23±7.32
Mean±SD*				87.05±9.87				55.94±10.97

Table VIII. Confusion counts and metrics for every spatial and DCT K=4096 run (validation split, from metrics.csv). †Not in metrics.csv; taken from the authors' experiment log. Sens., Spec., F1, BA in %.

Bench.	Method	Seed	TP	TN	FP	FN	Sens.	Spec.	F1	BA
B1	Spatial	42	12	68	0	8	60.00	100.00	75.00	80.00
B1	Spatial	123	9	68	0	11	45.00	100.00	62.07	72.50
B1	Spatial	2024	5	68	0	15	25.00	100.00	40.00	62.50
B1	DCT	42	0	68	0	20	0.00	100.00	0.00	50.00
B1	DCT	123	20	0	68	0	100.00	0.00	37.04	50.00
B1	DCT	2024	0	68	0	20	0.00	100.00	0.00	50.00
B2	Spatial	42	32	1057	0	3	91.43	100.00	95.52	95.71
B2	Spatial	123	35	1048	9	0	100.00	99.15	88.61	99.57
B2	Spatial	2024	35	1057	0	0	100.00	100.00	100.00	100.00
B2	DCT	42	0	1057	0	35	0.00	100.00	0.00	50.00
B2	DCT	123	0	1057	0	35	0.00	100.00	0.00	50.00
B2	DCT	2024	0	1057	0	35	0.00	100.00	0.00	50.00
B3	Spatial	42	177	929	0	5	97.25	100.00	98.61	98.63
B3	Spatial	123	181	925	4	1	99.45	99.57	98.64	99.51
B3	Spatial	2024	82	929	0	100	45.05	100.00	62.12	72.53
B3	DCT	42	0	929	0	182	0.00	100.00	0.00	50.00
B3	DCT	123	0	929	0	182	0.00	100.00	0.00	50.00
B3	DCT	2024	2	929	0	180	1.10	100.00	2.17	50.55
B4	Spatial	42	13	877	14	6	68.42	98.43	56.52	83.42
B4	Spatial	123	14	886	5	5	73.68	99.44	73.68	86.56
B4	Spatial†	2024	16	761	130	3	84.21	85.41	19.39	84.81
B4	DCT	42	17	826	65	2	89.47	92.70	33.66	91.09
B4	DCT	123	0	891	0	19	0.00	100.00	0.00	50.00
B4	DCT	2024	16	761	130	3	84.21	85.41	19.39	84.81
B5	Spatial	42	3	544	0	2	60.00	100.00	75.00	80.00
B5	Spatial	123	5	544	0	0	100.00	100.00	100.00	100.00
B5	Spatial	2024	4	544	0	1	80.00	100.00	88.89	90.00
B5	DCT	42	0	544	0	5	0.00	100.00	0.00	50.00
B5	DCT	123	5	138	406	0	100.00	25.37	2.40	62.68
B5	DCT	2024	0	544	0	5	0.00	100.00	0.00	50.00

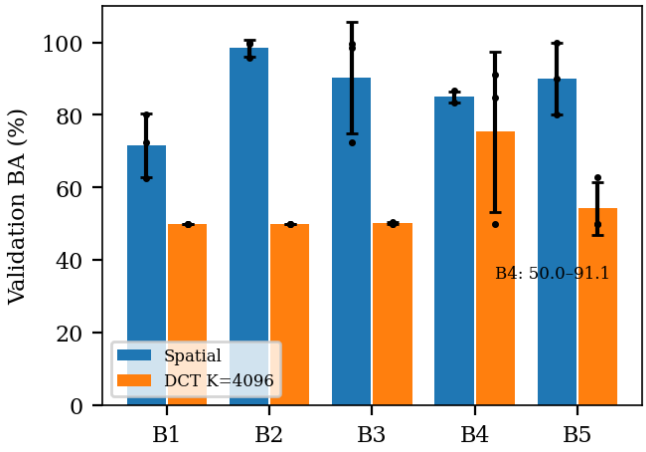


Fig. 7. Validation BA (mean ± SD, bars; individual seeds, dots) for the spatial baseline and DCT $K = 4096$. DCT seed spread on B4 spans 50.0% to 91.1%.

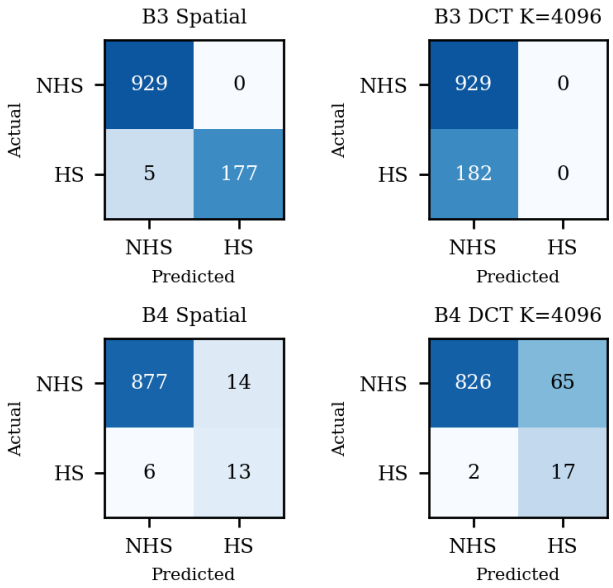


Fig. 8. Seed-42 validation confusion matrices (log-scaled shading, exact counts from metrics.csv) for B3 and B4. HS: hotspot; NHS: non-hotspot.

F. Statistical Comparison

A paired t-test across the five benchmark-level means indicated a significant difference ($t(4) = 4.51$, $p = 0.0107$), with the spatial representation performing better. The exact Wilcoxon signed-rank test yielded $p = 0.0625$, and the comparison rests on only five benchmark-level pairs, so the statistical evidence should be interpreted cautiously. All five paired differences favor the spatial baseline, the most extreme outcome possible for $n = 5$; the smallest attainable two-sided exact Wilcoxon p-value is 0.0625, so that test cannot reach 0.05 at this sample size regardless of effect size.

Table IX. Statistical comparison at benchmark level (n = 5).

Quantity	Value
Spatial mean±SD (n=5)	87.05±9.87%
DCT K=4096 mean±SD	55.94±10.97%
Mean difference	+31.11 pp
Paired t-test	$t(4) = 4.51$, $p = 0.0107$
95% CI of difference	+11.95 to +50.26 pp
Exact Wilcoxon signed-rank	$p = 0.0625$
Per-benchmark differences	+21.67, +48.43, +40.04, +9.63, +35.77

Table X. Comparison with related work.

Work	Representation, model, task	Difference from this study
Yang *et al.* [2, 3]	Block-wise DCT, zig-zag truncation, feature tensor; CNN with biased learning; ICCAD-12	Per-block coefficients preserve layout; imbalance handled by biased learning. Here: whole-image and single-size block variants, plain CNN; not comparable.
Xu *et al.* [9]	DCT inputs with learned channel selection; ResNet-50, MobileNetV2; ImageNet, COCO	Different task and architecture; selection here is fixed (zig-zag or row-major), not learned.
Verma *et al.* [12]	Spatial lightweight CNN, 12,873 parameters; ICCAD-12	Baseline source; reports 97.9% best (B2) and 95.3% average BA under its own protocol, not comparable to validation-split values here.

VIII. Discussion

Under the evaluated configuration, truncated DCT did not give a robust gain over the spatial input: the mean benchmark-level gap was large, the $K = 4096$ runs were mostly constant classifiers, and the few useful runs did not recur across seeds. The pattern is consistent with a model that finds no stable decision boundary from the DCT input within 5–10 epochs on imbalanced data and falls back on the majority class (or, once, the minority class). Direct tests of this reading were not performed.

Several observations constrain interpretation. First, full-image DCT behaved differently from $K = 4096$ (Table IV versus Table VII): it scored 87.56% on B2 and 91.33% on B3, where $K = 4096$ stayed at 50%, so conclusions about one cannot be transferred to the other. Second, the absence of a monotonic trend in K and the benchmark-specific optima suggest that no single K or selection rule is appropriate across B1–B5. Third, the zig-zag versus row-major contrast is single-seed, and given the seed spread seen on B4 it could partly reflect initialization.

Possible contributing factors, offered as hypotheses and not tested here, include loss of the spatial locality that 3×3 convolutions assume when a global transform is applied; information removed by the log and per-image min–max steps; the very small validation hotspot counts on B1 and B5; and the short training schedule. Block-wise DCT in the spirit of [2] preserves spatial structure and could behave differently, but the two block configurations screened here are insufficient to say so.

IX. Comparison with Related Frequency-Domain Work

No numerical superiority is claimed over any of these works because protocols differ in evaluation partition, preprocessing

and training schedule. Relative to [2, 3], the present networks receive a single resized, image-sized DCT map instead of a tensor of per-block coefficient channels, and no imbalance-specific training is used; the weak $K = 4096$ results should therefore not be read as evidence against the published block-DCT detector.

X. Limitations

(1) Only five ICCAD-12 benchmarks were used, and (2) the task is highly imbalanced, leaving very few validation hotspots on B1 (20) and B5 (5). (3) Robustness rests on three training seeds, (4) with a single fixed train/validation split. (5) Block-DCT screening covered two block sizes on B1 only because of computational cost. (6) Only the listed DCT preprocessing variants were tested, and (7) no claim is made that other frequency-domain representations are ineffective. (8) The statistical comparison has five benchmark-level pairs. (9) Other coefficient-selection strategies may behave differently. (10) Hardware, power-performance-area and runtime implications were not evaluated. In addition, results are for validation splits only; official-test performance is not reported in this experimental campaign.

XI. Future Work

Not yet performed, and suggested as next steps: a larger search over coefficient-selection rules; adaptive or learned frequency-band selection [9]; hybrid spatial–frequency architectures and multi-scale DCT; a systematic block-DCT study across all benchmarks; more seeds and cross-validation with stronger statistical testing; imbalance-aware training for DCT inputs; evaluation on the held-out official test partitions; and runtime and hardware-cost analysis of frequency-domain preprocessing.

XII. Conclusion

Under a fixed 80/20 split, three seeds and a 12,873-parameter CNN, truncated DCT with zig-zag selection at $K = 4096$ did not provide a robust improvement over the spatial representation on ICCAD-12 validation data. Outcomes depended on coefficient count, selection order and benchmark, and were seed-sensitive, most visibly on B4. The initial block-DCT screening was uninformative at 50% BA. The benchmark-level statistics favor the spatial baseline but are limited by five pairs and by disagreement between the t-test and the Wilcoxon test. These are findings about the evaluated design space, not about frequency-domain methods in general.

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